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(54) **HINGE, FLEXIBLE DISPLAY PANEL AND ELECTRONIC DEVICE**

(71) Applicant: **WUHAN CHINA STAR OPTOELECTRONICS SEMICONDUCTOR DISPLAY TECHNOLOGY CO., LTD.**, Wuhan (CN)

(72) Inventor: **Guobao Jiang**, Wuhan (CN)

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G06F 9/301; G06F 9/30; H05K 5/0226;
F16C 11/04; F16C 11/10
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

2019/0075196	A1 *	3/2019	Mok		G06F 1/1652
2021/0165466	A1 *	6/2021	Kang		G06F 1/1616
2022/0159109	A1 *	5/2022	Kang		G06F 1/1616

(Continued)

FOREIGN PATENT DOCUMENTS

CN	103576775	A	2/2014
CN	208919053	U	5/2019
CN	110442196	A	11/2019

(Continued)

OTHER PUBLICATIONS

International Search Report in International application No. PCT/CN2021/112503, mailed on Apr. 6, 2022.

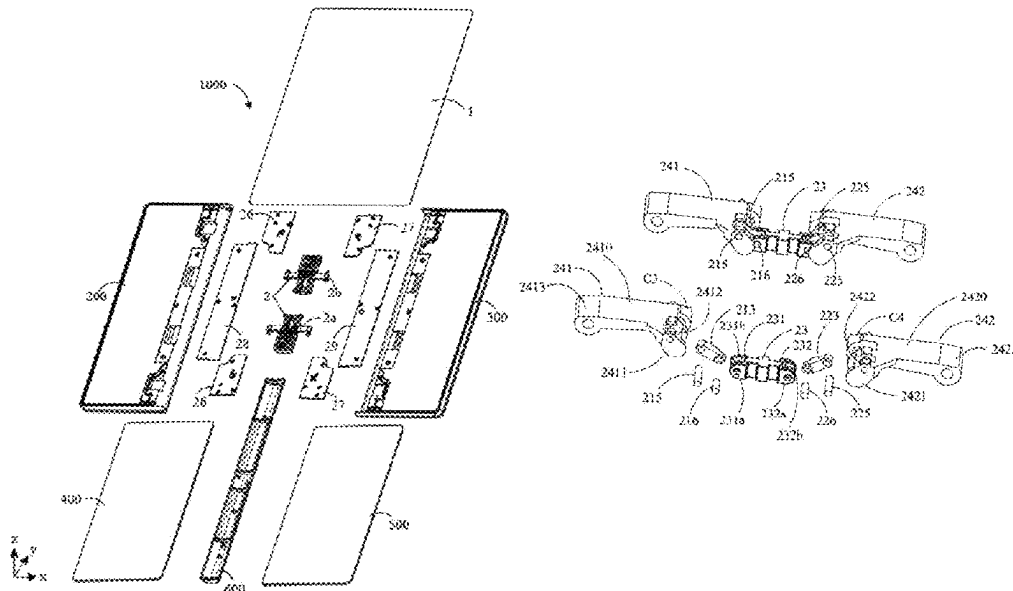
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Primary Examiner — Abhishek M Rathod

(57) **ABSTRACT**

A hinge, a flexible display panel, and an electronic device are provided by the present disclosure. The electronic device includes a flexible display panel which includes a hinge. The hinge includes a first rotation mechanism, a second rotation mechanism, and a moving block. The first rotation mechanism and the second rotation mechanism are interlinked by the moving block, so that the first rotation mechanism and the second rotation mechanism always maintain the same rotation speed, thereby achieving an effect of bilateral synchronous movement in the electronic device.

17 Claims, 11 Drawing Sheets



(56)

References Cited

U.S. PATENT DOCUMENTS

2023/0118440 A1 * 4/2023 Lee G06F 1/1652
361/807

FOREIGN PATENT DOCUMENTS

CN	211123830 U	7/2020
CN	112901643 A	6/2021
CN	113027899 A	6/2021
CN	113202857 A	8/2021
KR	20200047253 A	5/2020

OTHER PUBLICATIONS

Written Opinion of the International Search Authority in International application No. PCT/CN2021/112503, mailed on Apr. 6, 2022.

* cited by examiner

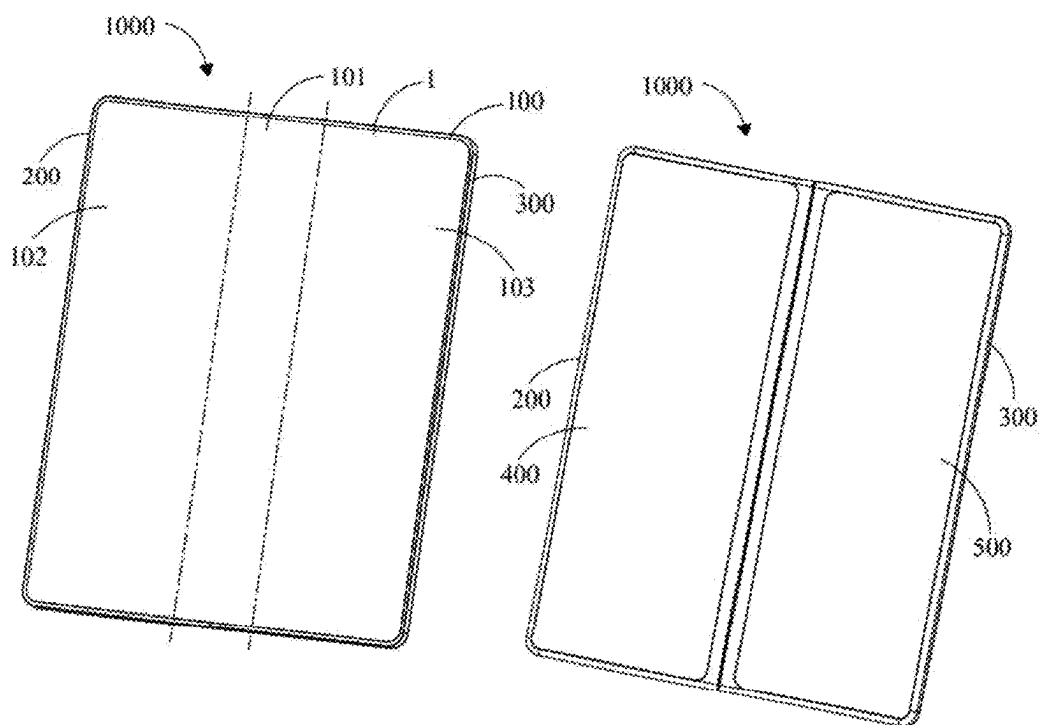


FIG. 1

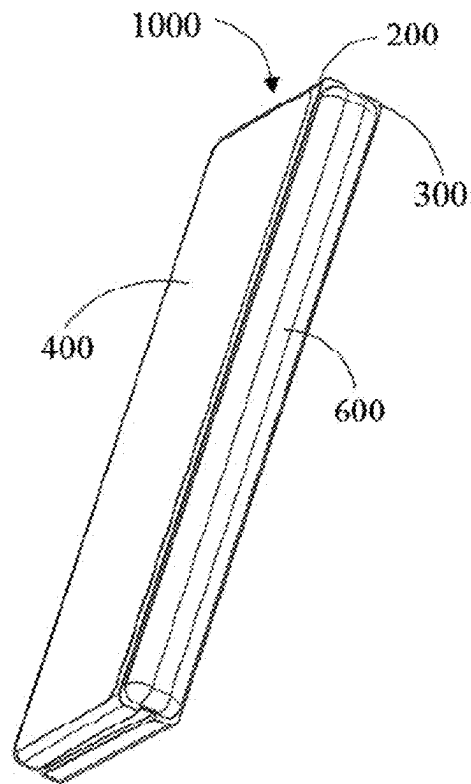


FIG. 2

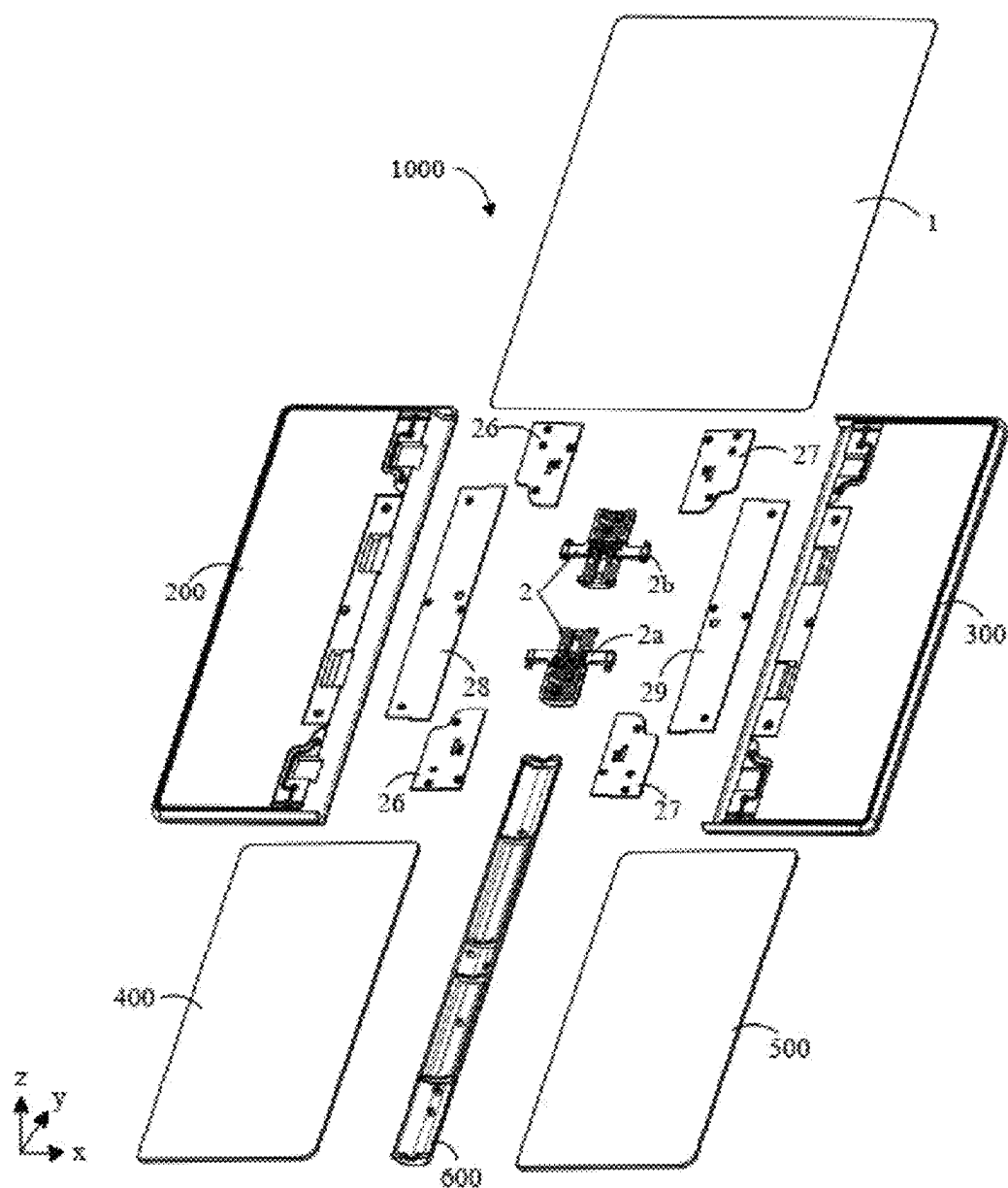


FIG. 3

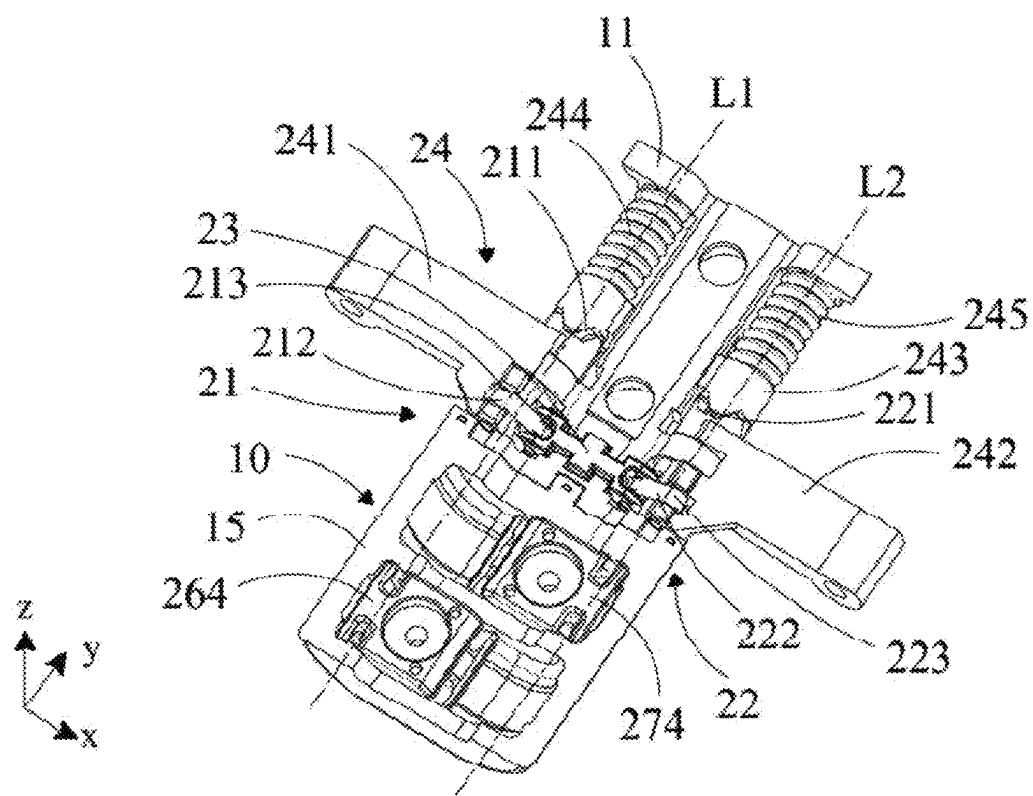


FIG. 4

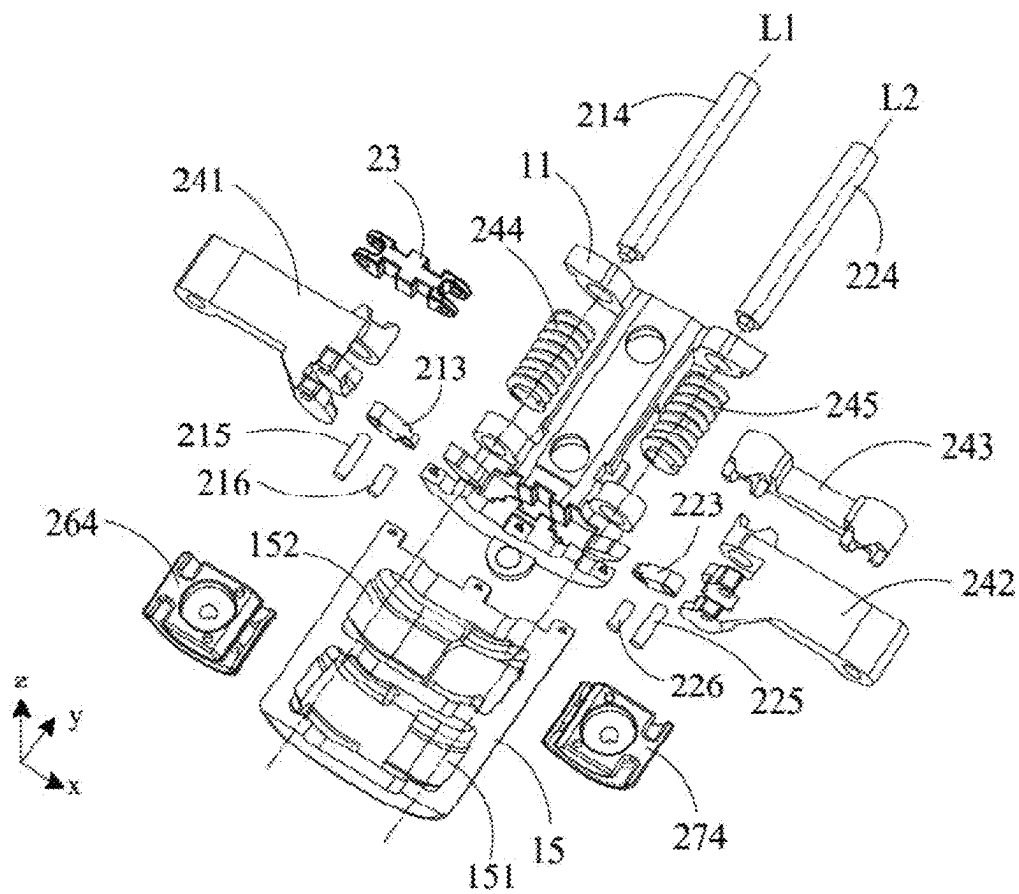


FIG. 5

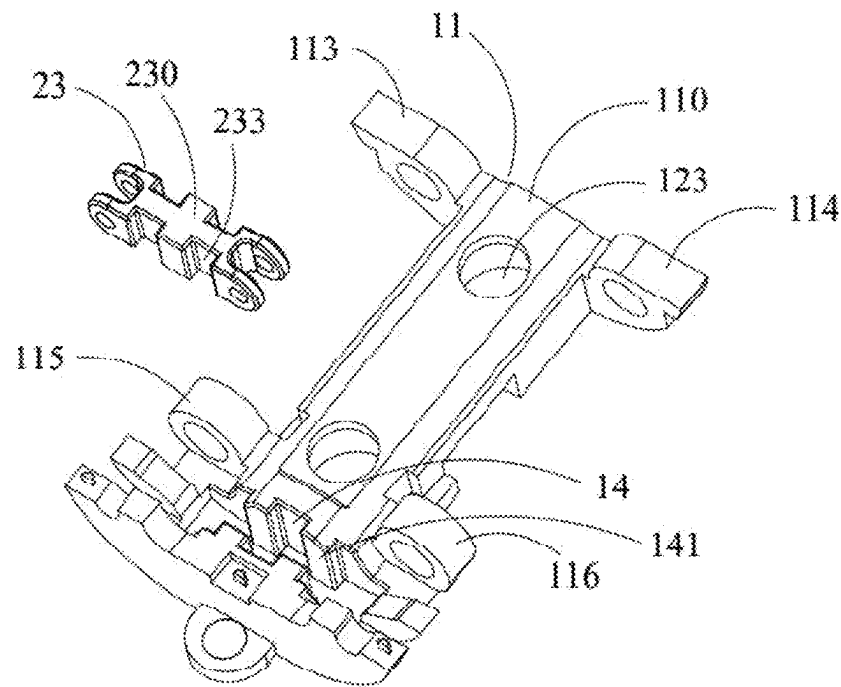


FIG. 6

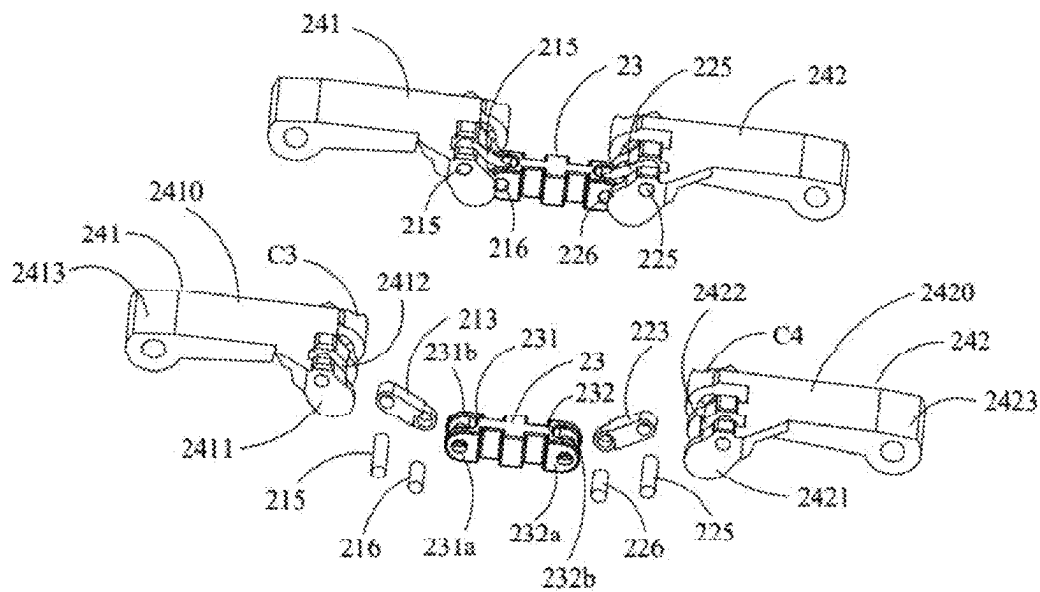


FIG. 7

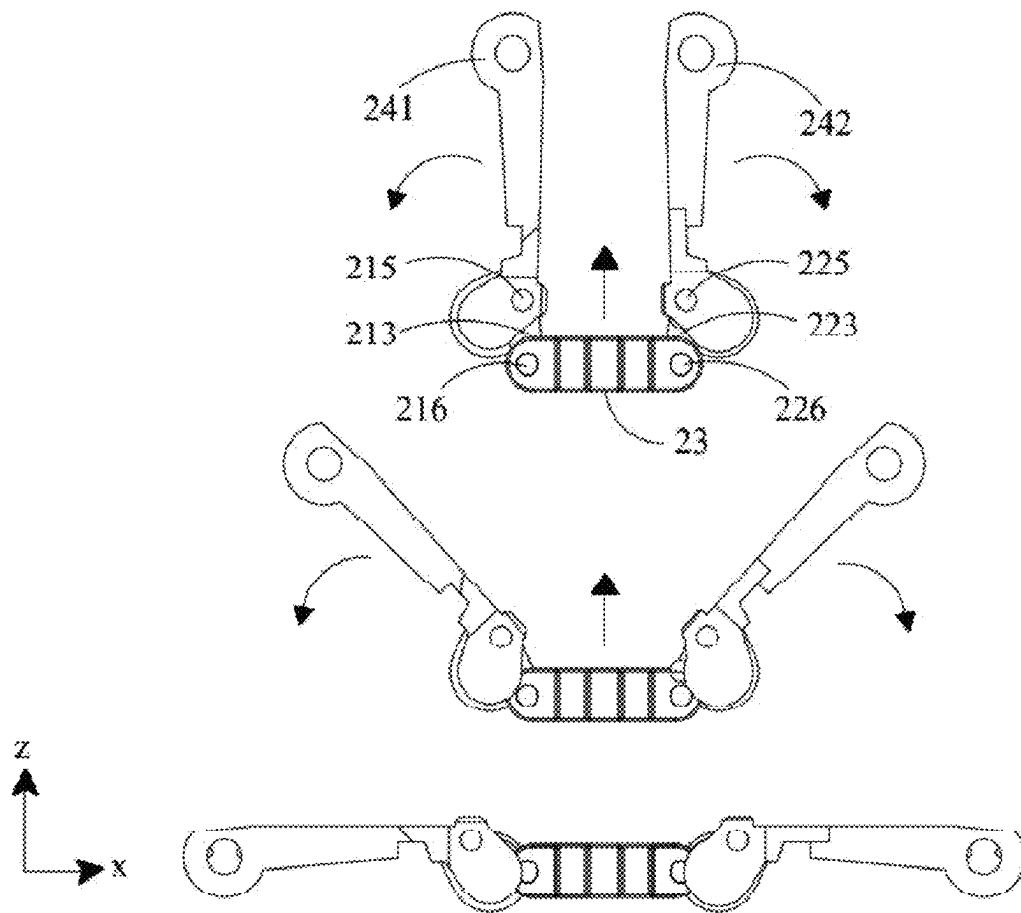


FIG. 8

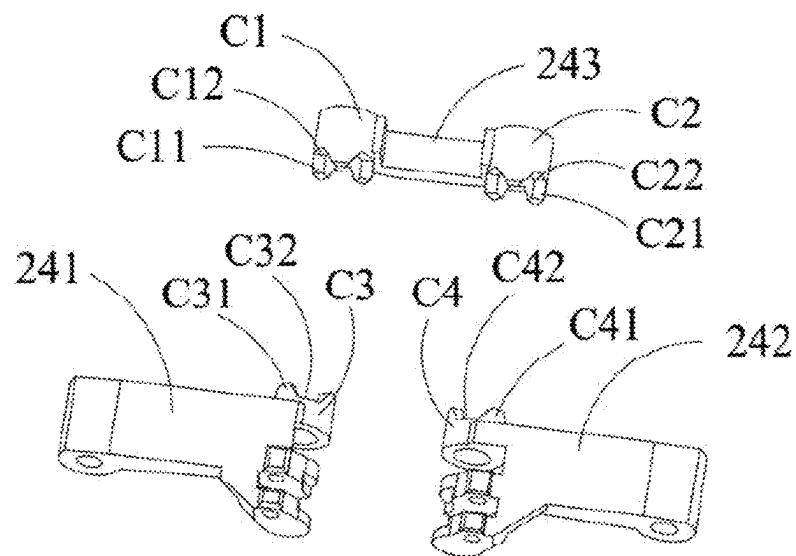


FIG. 9

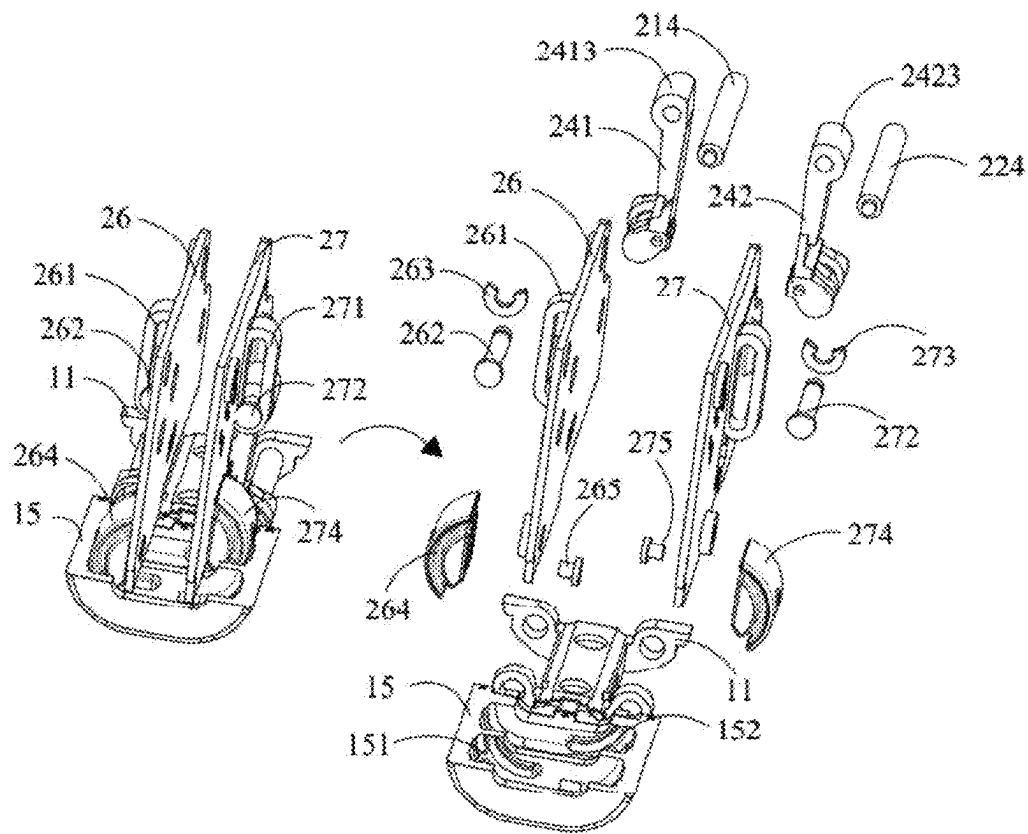


FIG. 10

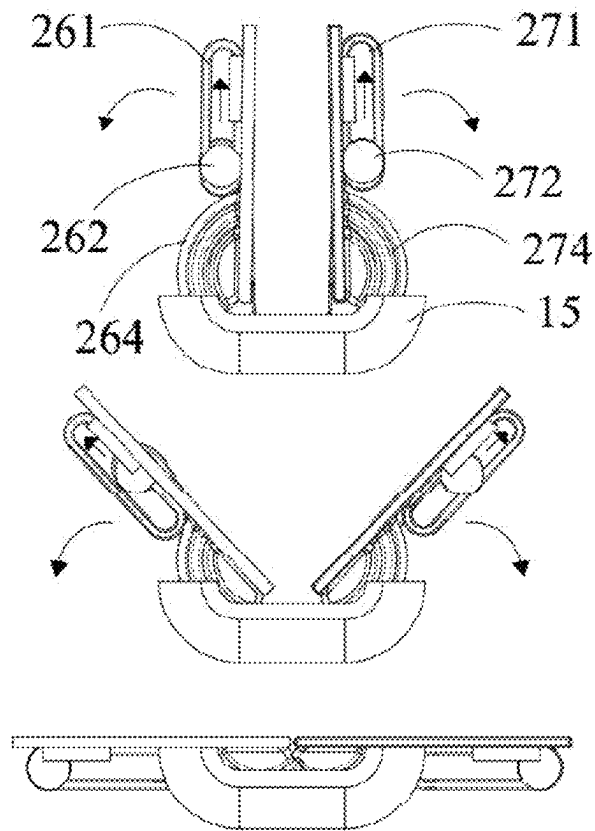


FIG. 11

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HINGE, FLEXIBLE DISPLAY PANEL AND ELECTRONIC DEVICE

FIELD OF INVENTION

The present disclosure relates to the field of display technologies, and in particular to a hinge, a flexible display panel and an electronic device.

BACKGROUND OF INVENTION

With continuous developments of display technologies, in order to meet different usage requirements, various displays with different characteristics have emerged. A development trend of intelligence, portability, and flexibility is one of the main development directions of current electronic devices. Compared with traditional flat panel display technologies, the most prominent advantages of flexible display technologies are breaking through an inherent concept of original two-dimensional display and expanding an application field of the display technologies to more portable electronic devices. A user can fold or roll an electronic device provided with the flexible display panel as needed, so as to reduce a size of the electronic device, thereby improving portability of the electronic device.

Technical Problems

The existing foldable electronic devices realize an effect of unfolding and folding by two rotation mechanisms disposed opposite to each other in the hinge. During folding or unfolding processes of the electronic devices, since the two rotation mechanisms in the hinge cannot always rotate at the same speed, a bilateral synchronous rotation cannot be achieved in the electronic devices, resulting in a flexible display panel main body of the electronic device being pressed or pulled by the hinge, thereby reducing service lives of the electronic devices.

As mentioned above, the hinge in the existing foldable electronic devices exists a problem that the bilateral synchronous rotation cannot be achieved. Therefore, it is necessary to provide a display panel and a display device to improve this defect.

SUMMARY OF INVENTION

Technical Solutions

Embodiments of the present disclosure provide a hinge, a flexible display panel, and an electronic device, which are used to solve a problem that a hinge in the existing foldable electronic device cannot realize bilateral synchronous rotation.

An embodiment of the present disclosure provides a hinge, including:

a fixed support;

a first rotation mechanism including a first rotation member and a first transmission member, wherein the first rotation member is rotatably connected to the fixed support, and the first transmission member is fixedly connected to the first rotation member;

a second rotation mechanism disposed opposite the first rotation mechanism and including a second rotation member and a second transmission member, wherein the second rotation member is rotatably connected to the fixed support, and the second transmission member is fixedly connected to the second rotation member; and

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a moving block, wherein the first transmission member and the second transmission member are rotatably connected to the moving block, so as to drive the moving block away from or close to the fixed support.

According to an embodiment of the present disclosure, the first transmission member includes a first link, the second transmission member includes a second link, and the moving block includes a base portion and a first extension portion and a second extension portion, which are connected to opposite sides of the base portion, wherein each of the first extension portion and the second extension portion includes two opposite extension plates extending from the base portion;

wherein one end of the first link is fixedly connected to the first rotation member, the other end of the first link is rotatably connected between the two extension plates of the first extension portion of the moving block, one end of the second link is fixedly connected to the second rotation member, and the other end of the second link is rotatably connected between the two extension plates of the second extension portion of the moving block.

According to an embodiment of the present disclosure, a first chute is defined by the fixed support, and the moving block is slidably disposed in the first chute.

According to an embodiment of the present disclosure, a wall surface of the first chute is provided with a protrusion, a groove is defined by the base portion of the moving block, and the protrusion is slidably received in the groove.

According to an embodiment of the present disclosure, the hinge further comprises a holding mechanism, and the holding mechanism includes a first holding arm, a second holding arm, and a first holding member, wherein the first holding member is provided with a first protrusion and a second protrusion, a first groove is defined by the first holding arm, and a second groove is defined by the second holding arm;

wherein the first holding arm is connected to the first rotation member, the second holding arm is connected to the second rotation member, and the first rotation member and the second rotation member are rotatably connected to the first holding member;

in response to the hinge being in one of a folded state or an unfolded state, the first protrusion is received in the first groove, and the second protrusion is received in the second groove; in response to the hinge being in the other state of the folded state or the unfolded state, the first protrusion is separated from the first groove, and the second protrusion is separated from the second groove.

According to an embodiment of the present disclosure, the holding mechanism includes a first elastic member and a second elastic member, and the fixed support includes a main body portion and a first connection portion and a second connection portion, which extend from opposite sides of the main body portion, wherein the first rotation member passes through the first connection portion, and the second rotation member passes through the second connection portion;

wherein the first elastic member surrounds the first rotation member and is clamped between the first holding member and the first connection portion, and the second elastic member surrounds the second rotation member and is clamped between the first holding member and the first connection portion.

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According to an embodiment of the present disclosure, the hinge further comprises a first support plate and a second support plate;

wherein the first support plate is connected to the first holding arm, so as to move under driving of the first holding arm, and the second support plate is connected to the second holding arm, so as to move under driving of the second holding arm.

According to an embodiment of the present disclosure, the hinge further comprises a first fixing member and a second fixing member, wherein a second chute is defined by the first support plate, a third chute is defined by the second support plate, the first fixing member passes through the second chute and is fixedly connected to the first holding arm, and the second fixing member passes through the third chute and is fixedly connected to the second holding arm;

wherein a position of the first fixing member in the second chute where the hinge is in the unfolded state is different from a position of the first fixing member in the second chute where the hinge being in the folded state; and a position of the second fixing member in the third chute where the hinge is in the unfolded state is different from a position of the second fixing member in the third chute where the hinge being in the folded state.

An embodiment of the present disclosure further provides a flexible display panel, including:

a flexible display panel main body including a flexible portion and a first portion and a second portion, which are connected to opposite ends of the flexible portion; and

a hinge, including:

a fixed support;

a first rotation mechanism including a first rotation member and a first transmission member, wherein the first rotation member is rotatably connected to the fixed support, and the first transmission member is fixedly connected to the first rotation member;

a second rotation mechanism disposed opposite the first rotation mechanism and including a second rotation member and a second transmission member, wherein the second rotation member is rotatably connected to the fixed support, and the second transmission member is fixedly connected to the second rotation member; and

a moving block, wherein the first transmission member and the second transmission member are rotatably connected to the moving block, so as to drive the moving block away from or close to the fixed support; wherein the first portion is fixedly connected to the first rotation member, the second portion is fixedly connected to the second rotation member, and the moving block is located under the flexible portion.

According to an embodiment of the present disclosure, the first transmission member includes a first link, the second transmission member includes a second link, and the moving block includes a base portion and a first extension portion and a second extension portion, which are connected to opposite sides of the base portion, wherein each of the first extension portion and the second extension portion includes two opposite extension plates extending from the base portion;

wherein one end of the first link is fixedly connected to the first rotation member, the other end of the first link is rotatably connected between the two extension plates of the first extension portion of the moving block, one end of the second link is fixedly connected to the second rotation member, and the other end of the

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second link is rotatably connected between the two extension plates of the second extension portion of the moving block.

According to an embodiment of the present disclosure, a first chute is defined by the fixed support, and the moving block is slidably disposed in the first chute.

According to an embodiment of the present disclosure, a wall surface of the first chute is provided with a protrusion, a groove is defined by the base portion of the moving block, and the protrusion is slidably received in the groove.

According to an embodiment of the present disclosure, the hinge further comprises a holding mechanism, and the holding mechanism includes a first holding arm, a second holding arm, and a first holding member, wherein the first holding member is provided with a first protrusion and a second protrusion, a first groove is defined by the first holding arm, and a second groove is defined by the second holding arm;

wherein the first holding arm is connected to the first rotation member, the second holding arm is connected to the second rotation member, and the first rotation member and the second rotation member are rotatably connected to the first holding member;

in response to the hinge being in one of a folded state or an unfolded state, the first protrusion is received in the first groove, and the second protrusion is received in the second groove; in response to the hinge being in the other state of the folded state or the unfolded state, the first protrusion is separated from the first groove, and the second protrusion is separated from the second groove.

According to an embodiment of the present disclosure, the holding mechanism includes a first elastic member and a second elastic member, and the fixed support includes a main body portion and a first connection portion and a second connection portion, which extend from opposite sides of the main body portion, wherein the first rotation member passes through the first connection portion, and the second rotation member passes through the second connection portion;

wherein the first elastic member surrounds the first rotation member and is clamped between the first holding member and the first connection portion, and the second elastic member surrounds the second rotation member and is clamped between the first holding member and the first connection portion.

According to an embodiment of the present disclosure, the hinge further comprises a first support plate and a second support plate;

wherein the first support plate is connected to the first holding arm, so as to move under driving of the first holding arm, and the second support plate is connected to the second holding arm, so as to move under driving of the second holding arm.

According to an embodiment of the present disclosure, the hinge further comprises a first fixing member and a second fixing member, wherein a second chute is defined by the first support plate, a third chute is defined by the second support plate, the first fixing member passes through the second chute and is fixedly connected to the first holding arm, and the second fixing member passes through the third chute and is fixedly connected to the second holding arm;

wherein a position of the first fixing member in the second chute where the hinge is in the unfolded state, is different from a position of the first fixing member in the second chute where the hinge is in the folded state; and a position of the second fixing member in the third

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chute where the hinge is in the unfolded state, is different from a position of the second fixing member in the third chute where the hinge is in the folded state.

An embodiment of the present disclosure further provides an electronic device, comprising: a flexible display panel, and the flexible display panel includes:

- a fixed support;
- a first rotation mechanism including a first rotation member and a first transmission member, wherein the first rotation member is rotatably connected to the fixed support, and the first transmission member is fixedly connected to the first rotation member;
- a second rotation mechanism disposed opposite the first rotation mechanism and including a second rotation member and a second transmission member, wherein the second rotation member is rotatably connected to the fixed support, and the second transmission member is fixedly connected to the second rotation member; and
- a moving block, wherein the first transmission member and the second transmission member are rotatably connected to the moving block, so as to drive the moving block away from or close to the fixed support; wherein the first portion is fixedly connected to the first rotation member, the second portion is fixedly connected to the second rotation member, and the moving block is located under the flexible portion.

According to an embodiment of the present disclosure, the first transmission member includes a first link, the second transmission member includes a second link, and the moving block includes a base portion and a first extension portion and a second extension portion, which are connected to opposite sides of the base portion, wherein each of the first extension portion and the second extension portion includes two opposite extension plates extending from the base portion;

wherein one end of the first link is fixedly connected to the first rotation member, the other end of the first link is rotatably connected to between the two extension plates of the first extension portion of the moving block, one end of the second link is fixedly connected to the second rotation member, and the other end of the second link is rotatably connected between the two extension plates of the second extension portion of the moving block.

According to an embodiment of the present disclosure, a first chute is defined by the fixed support, and the moving block is slidably disposed in the first chute.

According to an embodiment of the present disclosure, a wall surface of the first chute is provided with a protrusion, a groove is defined by the base portion of the moving block, and the protrusion is slidably received in the groove.

Beneficial effect of the embodiments of the present disclosure:

The embodiments of the present disclosure provide a hinge, a flexible display panel, and an electronic device. The electronic device includes a flexible display panel which includes a flexible display panel body and a hinge. The flexible display panel body includes a flexible portion and a first portion and a second portion, which are connected to opposite ends of the flexible portion. The hinge includes a first rotation mechanism, a second rotation mechanism, and a moving block. The first rotation mechanism includes a first rotation member and a first transmission member. The first rotation member is rotatably connected to the fixed support, and the first transmission member is fixedly connected to the first rotation member. The second rotation mechanism is disposed opposite the first rotation mechanism, and includes

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a second rotation member and a second transmission member. The second rotation member is rotatably connected to the fixed support, and the second transmission member is fixedly connected to the second rotation member. The first portion is fixedly connected to the first rotation member, the second portion is fixedly connected to the second rotation member, the moving block is located under the flexible portion, and the first transmission member and the second transmission member are rotatably connected to the moving block, so as to drive the moving block away from or close to the fixed support. The first rotation mechanism and the second rotation mechanism are interlinked by the moving block, so that the first rotation mechanism and the second rotation mechanism rotate at the same speed, thereby achieving an effect of bilateral synchronous movement in the electronic device, and preventing the flexible display panel main body from being pressed or pulled during folding or unfolding processes of the electronic device.

DESCRIPTION OF DRAWINGS

In order to illustrate technical solutions in the embodiments or in the prior art more clearly, attached drawings required for describing the embodiments or prior art will be simply explained as below. Apparently, the attached drawings for the following description are only some embodiments of the present disclosure. Those skilled in the art also could derive other attached drawings from these attached drawings without making a creative work.

FIG. 1 is a schematic diagram of the front and back structures of an electronic device provided in an embodiment of the present disclosure in an unfolded state.

FIG. 2 is a schematic structural diagram of an electronic device in a folded state according to an embodiment of the present disclosure.

FIG. 3 is a schematic exploded diagram of an electronic device provided by an embodiment of the present disclosure.

FIG. 4 is a schematic structural diagram of a hinge provided by an embodiment of the present disclosure.

FIG. 5 is a schematic exploded diagram of a hinge provided by an embodiment of the present disclosure.

FIG. 6 is a schematic structural diagram of a moving block and a fixed support provided by an embodiment of the present disclosure.

FIG. 7 is a schematic diagram of structure of a first holding arm, a second holding arm, and a moving block provided by an embodiment of the present disclosure.

FIG. 8 is a schematic diagram of the unfolding process of the hinge provided by an embodiment of the present disclosure.

FIG. 9 is a schematic exploded diagram of a holding mechanism provided by an embodiment of the present disclosure.

FIG. 10 is a schematic diagram of structures of the first support plate and the second support plate provided by an embodiment of the present disclosure.

FIG. 11 is a schematic diagram of movements of the first support plate and the second support plate provided by an embodiment of the present disclosure.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

The description of the following embodiments refers to the attached drawings to illustrate specific embodiments that can be implemented in the present disclosure. The directional terms mentioned in the present disclosure, such as

“upper”, “lower”, “front”, “back”, “left”, “right”, “inner”, “outer”, and “lateral” merely refer to the directions of the attached drawings. Therefore, the directional terms are used to illustrate and understand the present disclosure, rather than to limit the present disclosure. In the drawings, units with similar structures are indicated by the same reference numerals.

The present disclosure is further described below with reference to the drawings and specific embodiments.

An embodiment of the present disclosure provides a hinge 2, a flexible display panel 100, and an electronic device 1000. The electronic device 1000 includes the flexible display panel 100 which includes the hinge 2.

In the embodiment of the present disclosure, the electronic device 1000 may be a mobile terminal, such as a smart phone, a tablet computer, a notebook computer, and the like. The electronic device 1000 may further be a wearable terminal, such as a smart watch, a smart bracelet, smart glasses, an augmented reality device, and the like. An electronic device 1000 may further be a fixed terminal, such as a desktop computer, a TV, and the like.

As shown in FIGS. 1 to 3, FIG. 1 is a schematic structural diagram of the front and back of the electronic device in an unfolded state provided by an embodiment of the present disclosure, FIG. 2 is a schematic structural diagram of the electronic device in a folded state provided by an embodiment of the present disclosure, and FIG. 3 is a schematic exploded diagram of an electronic device provided by an embodiment of the present disclosure. The electronic device 1000 includes a flexible display panel 100, a first housing 200, a second housing 300, a first back plate 400, and a second back plate 500, and a third housing 600.

The flexible display panel 100 includes a flexible display panel main body 1. The flexible display panel main body 1 includes a flexible portion 101 and a first portion 102 and a second portion 103 located on opposite sides of the flexible portion 101. The flexible portion 101 is a folded portion of the flexible display panel main body 1. The first portion 102 may be disposed on at least a portion of the front side of the first housing 200, and the second portion 103 may be disposed on at least a portion of the front side of the second housing 300.

The first back plate 400 is disposed on a side of the first housing 200 away from the flexible display panel 100, and the second back plate 500 is disposed on a side of the second housing 300 away from the flexible display panel 100. The first housing 200 and the second housing 300 may individually have a separate space, which can receive electronic components in the electronic device, such as a motherboard, a printed circuit board, a battery, and the like, which are required for driving the flexible display panel 100.

It should be noted that the electronic device 1000 may have at least one rotation axis for folding the electronic device 1000. In the following description, a first direction x represents a direction perpendicular to the rotation axis, a second direction y represents a direction parallel to the rotation axis, and a third direction z represents a thickness direction of the electronic device 1000. A display surface of the electronic device may be defined by the first direction x and the second direction y.

The flexible display panel 100 further includes a hinge 2 which includes a first hinge 2a and a second hinge 2b, and the first hinge 2a and the second hinge 2b are individually disposed on opposite ends of the third housing 600. The first hinge 2a and the second hinge 2b may be disposed between the first housing 200 and the second housing 300. The first hinge 2a and the second hinge 2b are both connected to the

first housing 200 and the second housing 300. The first hinge 2a and the second hinge 2b may be configured to rotate the first housing 200 and the second housing 300 about two rotation axes parallel to the second direction y and are spaced in the first direction x.

In the embodiment of the present disclosure, structures of the first hinge 2a and the second hinge 2b are the same, and movement processes of the first hinge 2a and the second hinge 2b are also the same. In the following description, each schematic diagram merely takes the structure of the first hinge 2a as an example for illustration, and will not be repeated here. In practical applications, structures of the first hinge 2a and the second hinge 2b may further be different, but must be satisfied with the requirement that the first housing 200 and the second housing 300 can be rotated synchronously by the first hinge 2a and the second hinge 2b.

It should be noted that in the following description, the unfolded state may refer to the electronic device 1000 being completely unfolded, and a front side of the first housing 200 and a front side of the second housing 300 may form together a plane, so that a state of an angle of about 180 degrees is formed between the front side of the first housing 200 and the front side of the second housing 300. The folded state includes a completely folded state and a transitional folded state. The completely folded state may refer to the electronic device 1000 being completely folded. The front side of the first housing 200 and the front side of the second housing 300 may face each other, so that a state of an angle of about 0 degrees is formed between the front side of the first housing 200 and the front side of the second housing 300. The transitional folded state may refer to any transitional state between the unfolded state and the completely folded state of the electronic device 1000. The front side of the first housing 200 and the front side of the second housing 300 form together an included angle approximately greater than 0 degrees and less than 180 degrees.

The third housing 600 may be disposed between the first housing 200 and the second housing 300, and the third housing 600 may cover an outside of the hinge 2 and may protect the hinge 2.

As the electronic device 1000 is unfolded from the folded state to the unfolded state, a space between the side of the first housing 200 and the side of the second housing 300 may gradually narrow, and an opposite side of the third housing 600 may be inserted into the first housing 200 and the second housing 300. In response to the electronic device 1000 being in a unfolded state, the third housing 600 may be covered by the first back plate 400 and the second back plate 500, and will not be exposed to the outside.

As the electronic device 1000 is folded from the unfolded state to the folded state, the space between the side of the first housing 200 and the side of the second housing 300 may gradually widen, and the third housing 600 may be gradually exposed by the first housing 200 and the second housing 300 are. In response to the electronic device 1000 being in a completely folded state, at least a portion of the third housing 600 may be exposed to the outside between the side of the first housing 200 and the side of the second housing 300, and the first back plate 400 and the second back plate 500 can still cover some remaining portions of the third housing 600.

Refer to FIGS. 3 and 4. FIG. 4 is a schematic structural diagram of the hinge provided by an embodiment of the present disclosure. It should be noted that in the embodiment of the present disclosure, the structures of the first hinge 2a and the second hinge 2b are the same. In FIG. 4, only the first hinge 2a is taken as an example for illustration. Both the

first hinge **2a** and the second hinge **2b** include a fixed support **10**, a first rotation mechanism **21**, a second rotation mechanism **22**, and a moving block **23**.

The first rotation mechanism **21** and the second rotation mechanism **22** are individually movably connected to the fixed support **10**, and the first rotation mechanism **21** is disposed opposite the second rotation mechanism **22**. The rotation axes include a first axis **L1** and a second axis **L2**, and the first axis **L1** and the second axis **L2** are parallel to the second direction **y** and are spaced in the first direction **x**. The first rotation mechanism **21** can rotate about the first axis **L1**, and the second rotation mechanism **22** can rotate about the second axis **L2**. In the embodiment of the present disclosure, the structures of the first rotation mechanism **21** and the second rotation mechanism **22** are the same. In practical applications, the structures of the first rotation mechanism **21** and the second rotation mechanism **22** may be the same or different, and are not limited thereto.

The moving block **23** is movably connected to the first rotation mechanism **21** and the second rotation mechanism **22**, and is located under the flexible portion **101** of the flexible display panel main body **1**. During the folding or unfolding process of the electronic device **1000**, the moving block **23** merely moves along the third direction **z** or a reverse movement opposite the third direction **z**, so that the first rotation mechanism **21** and the second rotation mechanism **22** can maintain a synchronous movement.

The first rotation mechanism **21** includes a first rotation member **211** and a first transmission member **212**. The first rotation member **211** is rotatably connected to the fixed support **10**, and the first transmission member **212** is fixedly connected to the first rotation member **211**. The second rotation mechanism **22** includes a second rotation member **221** and a second transmission member **222**, the second rotation member **221** is rotatably connected to the fixed support **10**, and the second transmission member **222** is fixedly connected to the second rotation member **221**. The first transmission member **212** and the second transmission member **222** are rotatably connected to the moving block **23**, so as to drive the moving block **23** away from or close to the fixed support **10**.

Refer to FIGS. 4 to 7. FIG. 5 is a schematic exploded view of a hinge provided by an embodiment of the present disclosure, FIG. 6 is a structural schematic view of a moving block and a fixed support provided by an embodiment of the present disclosure, and FIG. 7 is a structural schematic view of a first holding arm, a second holding arm and a moving block provided by an embodiment of the present disclosure.

The fixed support **10** includes a first fixed support **11** which includes a main body portion **110** and a first connection portion **113**, a second connection portion **114**, a third connection portion **115**, and a fourth connection portion **116**, which protrude from the main body portion **110**. Two through holes **123** are defined by the main body portion **110**, and the through holes **123** are used to fix the first fixed support **11** to the third housing **600** by a screw connection.

The first connection portion **113** and the second connection portion **114** are respectively located on opposite sides of the main body portion **110**, the first connection portion **113** and the third connection portion **115** are located on the same side of the main body portion **110**, and the second connection portion **114** and the fourth connection portion **116** are located on the same side of the main body portion **110**. A shaft hole is individually defined by the first connection portion **113**, the second connection portion **114**, the third connection portion **115**, and the fourth connection portion **116**.

Both the first hinge **2a** and the second hinge **2b** include a holding mechanism **24** which includes a first holding arm **241** and a second holding arm **242** disposed opposite the first holding arm **241**. The first holding arm **241** includes a first body portion **2410** and a fifth connection portion **2411** and a third cam **C3**, which protrude from the same side of the first body portion **2410**. The third cam **C3** is spaced from the fifth connection portion **2411**. A shaft hole is defined by the third cam **C3**, and a first avoidance groove **2412** is defined by the fifth connection portion **2411**. A shaft hole is individually defined by two side walls of the first avoidance groove **2412**.

The second holding arm **242** includes a second body portion **2420** and a sixth connection portion **2421** and a fourth cam **C4**, which protrude from the same side of the second body portion **2420**. The fourth cam **C4** is spaced from the sixth connection portion **2421**. A shaft hole is defined by the fourth cam **C4**, and a second avoidance groove **2422** is defined by the sixth connection portion **2421**. A shaft hole is individually defined by two side walls of the second avoidance groove **2422**.

The first transmission member **212** includes a first link **213**, the second transmission member **222** includes a second link **223**, and the moving block **23** includes a base portion **230** and a first extension portion **231** and a second extension portion **232**, which are connected to opposite sides of the base portion **230**. Both of the first extension portion **231** and the second extension portion **232** include two opposite extension plates extending from the base portion **230**. One end of the first link **213** is fixedly connected to the first rotation member **211**, and the other end of the first link **213** is rotatably connected between the two extension plates of the first extension portion **231** of the moving block **23**. One end of the second link **223** is fixedly connected to the second rotation member **221**, and the other end of the second link **223** is rotatably connected between the two extension plates of the second extension portion **232** of the moving block **23**.

The first extension portion **231** includes a first extension plate **231a** and second extension plate **231b** opposite the first extension plate **231a**, and the first extension plate **231a** and the second extension plate **231b** are spaced apart from each other. A coaxial shaft hole is individually defined by the first extension plate **231a** and the second extension plate **231b**. The second extension portion **232** includes a third extension plate **232a** and a fourth extension plate **232b** opposite the third extension plate **232a**, and the third extension plate **232a** and the fourth extension plate **232b** are spaced apart from each other. A coaxial shaft hole is individually defined by the third extension plate **232a** and the fourth extension plate **232b**.

The first rotation member **211** includes a first rotation shaft **214**, a third rotation shaft **215**, and a fifth rotation shaft **216**. An axis of the first rotation shaft **214** coincides with the first axis **L1**, and the first rotation shaft **214** passes through the first connection portion **113**, the third connection portion **115**, and the third cam **C3**. A shaft hole is individually defined by both ends of the first link **213**, a first end of the first link **213** extends into the first avoidance groove **2412** of the fifth connection portion **2411** of the first holding arm **241**, and the third rotation shaft **215** passes through the first end of the first link **213** and the fifth connection portion **2411** of the first holding arm **241**. The first holding arm **241** can rotate relative to the third rotation shaft **215**, and the first link **213** is fixedly connected to the third rotation shaft **215**. A second end of the first link **213** extends between the first extension plate **231a** and the second extension plate **231b** of the first extension portion **231** of the moving block **23**, and

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the fifth rotation shaft **216** passes through the second portion of the first link **213** and the first extension portion **231**. The first link **213** can rotate relative to the fifth rotation shaft **216**, and the moving block **23** is fixedly connected to the fifth rotation shaft **216**.

The second rotation member **221** includes a second rotation shaft **224**, a fourth rotation shaft **225**, and a sixth rotation shaft **226**. An axis of the second rotation shaft **224** coincides with the second axis **L2**. The second rotation shaft **224** passes through the second connection portion **114** and the fourth connection portion **116**. A shaft hole is individually defined by both ends of the second link **223**. A first end of the second link **223** extends into the second avoidance groove of the second holding arm **242**, and the fourth rotation shaft **225** passes through the first end of the second link **223** and the sixth connection portion **2421** of the second holding arm **242**. The sixth connection portion **2421** can rotate relative to the fourth rotation shaft **225**, and the second link **223** is fixedly connected to the fourth rotation shaft **225**. A second end of the second link **223** extends between the third extension plate **232a** and the fourth extension plate **232b** of the second extension portion **232** of the moving block **23**. The sixth rotation shaft **226** passes through the second end of the second link **223** and the second extension portion **232**. The second link **223** can rotate relative to the sixth rotation shaft **226**, and the moving block **23** is fixedly connected to the sixth rotation shaft **226**.

Refer to FIG. 4 and FIG. 8. FIG. 8 is a schematic diagram of an unfolding process of a hinge provided by an embodiment of the present disclosure. In the unfolding process of the hinge, the first holding arm **241** drives the third rotation shaft **215** to rotate around the first axis **L1** in a direction of the arrow as shown in the figure, the third rotation shaft **215** drives the first link **213** to rotate around the fifth rotation shaft **216** in the direction of the arrow as shown in the figure, the second holding arm **242** drives the fourth rotation shaft **225** to rotate around the second axis **L2** in the direction of the arrow as shown in the figure, and the fourth rotation shaft **225** drives the second link **223** to rotate around the sixth rotation shaft **226** in the direction of the arrow as shown in the figure. The moving block **23** is driven by the first link **213** and the second link **223** to move in the direction of the arrow shown in the figure (i.e., the third direction **z**).

It can be understood that the unfolding process of the hinge as shown in FIG. 8 is reversible. During the folding process of the hinge, the first holding arm **241** drives the third rotation shaft **215** to rotate around the first axis **L1** in a direction opposite the direction of the arrow shown in the figure, the third rotation shaft **215** drives the first link **213** to rotate around the fifth rotation shaft **216** in the direction opposite the direction of the arrow shown in the figure, the second holding arm **242** drives the fourth rotation shaft **225** to rotate around the second axis **L2** in the direction opposite the direction of the arrow shown in the figure, and the fourth rotation shaft **225** drives the second link **223** to rotate around the sixth rotation shaft **226** in the direction opposite the direction of the arrow shown in the figure. The moving block **23** is driven by the first link **213** and the second link **223** to move in the direction opposite the direction of the arrow as shown in the figure (i.e., a direction opposite the third direction **z**).

During the unfolding or folding process of the hinge, the moving block **23** always moves in the third direction **z** or the direction opposite the third direction **z**, and the first extension portion **231** and the second extension portion **232** of the moving block **23** always maintain in the same horizontal plane. In this way, the first link **213** and the second link **223**

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can respectively rotate around the fifth rotation shaft **216** and the sixth rotation shaft **226** at the same rotation speed, thereby achieving an effect of synchronous rotation of the first rotation mechanism **21** and the second rotation mechanism **22**.

A first chute **14** is defined by the fixed support **10**, and the moving block **23** is slidably disposed in the first chute **14**.

As shown in FIG. 6, a first chute **14** is defined by the first fixed support **11**, and the first chute **14** penetrates a main body portion **110** of the first fixed support **11** in the third direction **z**. The moving block **23** is slidably disposed in the first chute **14**, and can only move in the first chute **14** in the third direction **z** or a direction opposite the third direction **z**, so as to ensure that the first rotation mechanism **21** and the second rotation mechanism **22** can rotate synchronously, and stuck of the moving block **23** during the sliding process can be prevented.

A protrusion **141** is provided on a wall surface of the first chute **14**, a groove **233** is defined by the base portion **230** of the moving block **23**, and the protrusion **141** is slidably received in the groove **233**.

As shown in FIG. 6, two opposite side walls of the first chute **14** are individually provided with a protrusion **141**, and a groove **233** is individually defined by portions of opposite sides of the base portion **230** of the moving block **23** corresponding to the protrusions **141**. The protrusion **141** is slidably received in the groove **233**. In this way, a movement direction of the moving block **23** is further limited by the protrusion **141**, so as to prevent the moving block **23** from being deflected left and right, thereby reducing a virtual position of the moving block **23** matched with the first chute **14**, and increasing stability of the moving block **23** during moving process.

In practical applications, a number of the grooves **233** defined by the moving block **23** and a number of the protrusions **141** on the wall surface of the first chute **14** can be determined according to a size of the moving block **23** and a size of the first chute **14**. The larger the sizes of the moving block **23** and the first chute **14** are, the more the numbers of the required grooves **233** and the protrusions **141** are. The number of the grooves **233** defined by the moving block **23** and the number of the protrusions **141** on the wall surface of the first chute **14** can be determined according to actual situations, and may be 1, 2, 3, or more, etc., and is not limited here.

The holding mechanism **24** further includes a first holding member **243** which is provided with a first protrusion **C11** and a second protrusion **C21**. A first groove **C32** is defined by the first holding arm **241**, and a second groove **C42** is defined by the second holding arm **242**. The first holding arm **241** is connected to the first rotation member **211**, the second holding arm **242** is connected to the second rotation member **221**, and the first rotation member **211** and the second rotation member **221** are rotatably connected to the first holding member **243**. In response to the hinge being in one of the folded state or the unfolded state, the first protrusion **C11** is received in the first groove **C32**, and the second protrusion **C21** is received in the second groove **C42**. In response to the flexible portion **101** being in the other state of the folded state or the unfolded state, the first protrusion **C11** is separated from the first groove **C32**, and the second protrusion **C21** is separated from the second groove **C42**.

Refer to FIG. 4, FIG. 5, and FIG. 9. FIG. 9 is a schematic exploded view of a holding mechanism provided by the embodiment of the present disclosure. In an embodiment, a snap-fit slot is defined by a bottom surface of the main body portion **110** of the first fixed support **11**, and a main body

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portion of the first holding member **243** is snapped into the snap-fit slot. The first holding member **243** includes a first cam **C1** and a second cam **C2**, which protrude from the main body portion. The first cam **C1** may include a plurality of first protrusions **C11** which protrude from an edge of the first cam **C1** in the second direction **y**. A third groove **C12** is defined between the first protrusions **C11** adjacent to each other. The first rotation shaft **214** passes through the first cam **C1**, and can rotate relative to the first cam **C1**. The second cam **C2** may include a plurality of second protrusions **C21** which protrude from an edge of the second cam **C2** in the second direction **y**. A fourth groove **C21** is defined between the second protrusions **C21** adjacent to each other. The first rotation shaft **214** passes through the second cam **C2**, and can rotate relative to the second cam **C2**.

The third cam **C3** of the first holding arm **241** may include a plurality of third protrusions **C31** protruding from an edge of the third cam **C3** in the second direction **y**, and a first groove **C32** is defined between the third protrusions **C31** adjacent to each other. The first rotation shaft **214** passes through the third cam **C3**. The second holding arm **242** includes a fourth cam **C4** protruding from a body of the second holding arm **242** in the first direction **x**. The fourth cam **C4** may include a plurality of fourth protrusions **C41** protruding from an edge of the fourth cam **C4** in the second direction **y**. A second groove **C42** is defined by the fourth protrusions **C41** adjacent to each other, and the second rotation shaft **224** passes through the fourth cam **C4**.

Each of the first protrusion **C11** to the fourth protrusion **C41** has a flat end which has a sliding surface. The sliding surface is an abutment surface of two inclined side surfaces of each protrusion.

The holding mechanism **24** further includes a first elastic member **244** and a second elastic member **245**, and both the first elastic member **244** and the second elastic member **245** are springs. The first elastic member **244** surrounds the first rotation shaft **214**, and is clamped between the first holding member **243** and the first connection portion **113**. The second elastic member **245** surrounds the second rotation shaft **224**, and is clamped between the first holding member **243** and the second connection portion **114**.

In response to the electronic device **1000** being in an unfolded state or a completely folded state, the hinge is in an unfolded state or a completely folded state, the flexible portion **101** is in an unfolded state or a completely folded state, the first cam **C1** of the first holding member **243** is ordinarily engaged with the third cam **C3** of the first holding arm **241**, and the second cam **C2** of the first holding member **243** is ordinarily engaged with the fourth cam **C4** of the second holding arm **242**.

The ordinary engagement means that, for example, the first protrusion **C11** of the first cam **C1** is received in the first groove **C32** of the third cam **C3**; the third protrusion **C31** is received in the third groove **C12**; the second protrusion **C21** of the second cam **C2** is received in the second groove **C42** of the fourth cam **C4**; and the fourth protrusion **C41** is received in the fourth groove **C22**.

In response to the electronic device **1000** being in the transitional folded state, the hinge is in the transitional folded state, the flexible portion **101** is in the folded state, the first cam **C1** of the first holding member **243** is in open engagement with the third cam **C3** of the first holding arm **241**, and the second cam **C2** of the first holding member **243** is in open engagement with the fourth cam **C4** of the second holding arm **242**.

The open engagement means that, for example, the first protrusion **C11** of the first cam **C1** is disengaged from the

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first groove **C32** of the third cam **C3**, and a sliding surface of the first protrusion **C11** is in contact with a sliding surface of the third protrusion **C31**, and the second protrusion **C21** of the second cam **C2** is disengaged from the second groove **C42** of the fourth cam **C4**, and a sliding surface of the second protrusion **C21** is in contact with a sliding surface of the fourth protrusion **C41**.

During the process of the electronic device **1000** being folded from the unfolded state to the transitional folded state, the first holding arm **241** rotates about the first axis **L1**, the second holding arm **242** rotates about the second axis **L2**, and the cams engaged with each other gradually changes from the ordinary engagement to the open engagement, where the first protrusion **C11** is gradually disengaged from the first groove **C32**, the second protrusion **C21** is gradually disengaged from the second groove **C42**, and the first elastic member **244** and the second elastic member **245** are compressed by the first holding member **243**, resulting in a gradual increase in an amount of elastic deformation. In response to an angle between the first portion **102** and the second portion **103** ranging from 0° to 30° , under an action of the compressing force of each of the elastic members and a pressure angle of each of the cams, the electronic device **1000** can be restored to an unfolded state without external force.

In response to the electronic device **1000** being in a transitional folded state, the plurality of cams engaged with each other are in the open engagement, and the pressure angle of each of the cams is 0° , the angle between the first portion **102** and the second portion **103** may be maintained at 150° , and the elastic deformation amount of each of the elastic member remains unchanged, so that the electronic device **1000** can be maintained in a transitional folded state without external force.

During the electronic device **1000** being folded from the transitional folded state to the completely folded state, the first protrusion **C11** is gradually received in the first groove **C32**, the second protrusion **C21** is gradually received in the second groove **C42**, the elastic deformation of the first elastic member **244** and the second elastic member **245** gradually decreases. In response to the angle between the first portion **102** and the second portion **103** of the flexible display panel body ranging from 150° to 180° , under the action of the compressing force of each of the elastic members and the pressure angle of the cams, the electronic device **1000** can automatically be folded from the transitional folded state to the completely folded state without external force.

In practical applications, a number of the holding members in the holding mechanism **24** is not limited to 1 or 2 mentioned in the above embodiments, but may also be 3, 4, or more. A number of the elastic members needs to meet the number of the holding members. The larger the size of the electronic device is, the greater the number of the required holding members is.

Refer to FIG. 10. FIG. 10 is a schematic diagram of structures of a first support plate and a second support plate provided by an embodiment of the present disclosure. Each of the first hinge **2a** and the second hinge **2b** includes a first support plate **26** and a second support plate **27**. The first support plate **26** is connected to the first holding arm **241**, so as to move under driving of the first holding arm **241**, and the second support plate **27** is connected to the second holding arm **242**, so as to move under driving of the second holding arm **242**. The first support plate **26** is fixedly connected to a front side of the first housing **200** by screws. The first support plate **26** can be used to support the first

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portion 102. The second support plate 27 is fixedly connected to a front side of the second housing 300 by screws. The second support plate 27 can be used to support the second portion 103.

A second chute 261 is defined by the first support plate 26, and a third chute 271 is defined by the second support plate 27. Both the second chute 261 and the third chute 271 are linear chutes. The first holding arm 241 includes a seventh connect portion 2413 protruding from one end of the first body portion 2410 opposite the fifth connection portion 2411. A shaft hole is defined by the seventh connection portion 2413. The second holding arm 242 includes an eighth connection portion 2423 protruding from one end of the second body portion 2420 opposite the sixth connection portion 2421. A shaft hole is defined by the eighth connection portion 2423.

The hinge further includes a first fixing member 262, a second fixing member 272, a third fixing clip 263, and a fourth fixing clip 273. The first fixing member 262 and the second fixing member 272 are both pin shafts. The first fixing member 262 passes through the second chute 261 and the shaft hole of the seventh connection portion 2413, and is fixedly connected to the seventh connection portion 2413. A third fixing clip 263 surrounds one end of the first fixing member 262 passing through seventh connect portion 2413, and the first fixing member 262 can slide and rotate relative to the second chute 261. The second fixing member 272 passes through the third chute 271 and the shaft hole of the eighth connection portion 2423, and is fixedly connected to the eighth connection portion 2423. A fourth fixing clip 273 surrounds one end of the second fixing member 272 passing through the eighth connection portion 2423, and the second fixing member 272 can slide and rotate relative to the third chute 271.

Both the first hinge 2a and the second hinge 2b include a first sliding block 264 and a second sliding block 274. The first sliding block 264 is fixedly connected to the first support plate 26 by a first screw 265, and the second sliding block 274 is fixedly connected to the second support plate 27 by a second screw 275.

The fixed support 10 includes a second fixed support 15 which is fixedly connected to the first fixed support 11 by a screw connection or a snap-fit manner. A fourth chute 151 and a fifth chute 152 are defined by the second fixed support 15, and the first sliding block 264 is slidably disposed in the fourth chute 151, and the second sliding block 274 is slidably disposed in the fifth chute 152.

A position of the first fixing member 262 in the second chute 261 where the hinge is in the unfolded state is different from a position of the first fixing member 262 in the second chute 261 where the hinge is in the folded state. A position of the second fixing member 272 in the third chute 271 where the hinge is in the unfolded state is different from a position of the second fixing member 272 in the third chute 271 where the hinge is in the folded state.

Refer to FIG. 11. FIG. 11 is a schematic diagram of movements of the first support plate and the second support plate provided by an embodiment of the present disclosure. In response to the electronic device 1000 being in a completely folded state, the hinge is in the folded state, the flexible portion 101 is in the folded state, the first fixing member 262 is located on one end of the second chute 261 close to the second fixed support 15, and the second fixing member 272 is located on one end of the third chute 271 close to the second fixed support 15.

During a process of the electronic device 1000 being unfolded from the completely folded state to the unfolded

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state, the first support plate 26 and the second support plate 27 rotate opposite to each other in a direction of an arrow as shown in the figure. During the process of the first support plate 26 rotating in the direction of the arrow as shown in the figure, the first fixing member 262 drives the first holding arm 241 to rotate in the same direction, slides in the second chute 261 in the direction of the arrow as shown in the figure, and gradually moves away from the second fixed support 15. During the process of the support plate 27 rotating in the direction of the arrow as shown in the figure, the second fixing member 272 drives the second holding arm 242 to rotate in the same direction, slides in the third chute 271 in the direction of the arrow as shown in the figure, and gradually moves away from the second fixed support 15.

In response to the electronic device 1000 being in the unfolded state, the hinge is in the unfolded state, the flexible portion 101 is in the unfolded state, the first fixing member 262 is located on one end of the second chute 261 away from the second fixed support 15, and the second fixing member 272 is located on one end of the third chute 271 away from the second fixed support 15.

In an embodiment, as shown in FIG. 3, the hinge further includes a third support plate 28 and a fourth support plate 29. The third support plate 28 is fixedly connected to the front side of the first housing 200 by screws, the third support plate 28 and the first support plate 26 are on the same plane, and are used to support the first portion 102. The fourth support plate 29 and the second housing 300 are fixedly connected by screws, the fourth support plate 29 and the second support plate 27 are on the same plane, and are used to support the second portion 103.

Beneficial effect of the embodiments of the present disclosure:

The embodiments of the present disclosure provide a hinge, a flexible display panel, and an electronic device. The electronic device includes a flexible display panel which includes a flexible display panel body and a hinge. The flexible display panel body includes a flexible portion and a first portion and a second portion, which are connected to opposite ends of the flexible portion. The hinge includes a first rotation mechanism, a second rotation mechanism, and a moving block. The first rotation mechanism includes a first rotation member and a first transmission member. The first rotation member is rotatably connected to the fixed support, and the first transmission member is fixedly connected to the first rotation member. The second rotation mechanism is disposed opposite the first rotation mechanism, and includes a second rotation member and a second transmission member. The second rotation member is rotatably connected to the fixed support, and the second transmission member is fixedly connected to the second rotation member. The first portion is fixedly connected to the first rotation member, the second portion is fixedly connected to the second rotation member, the moving block is located under the flexible portion, and the first transmission member and the second transmission member are rotatably connected to the moving block, so as to drive the moving block away from or close to the fixed support. The first rotation mechanism and the second rotation mechanism are interlinked by the moving block, so that the first rotation mechanism and the second rotation mechanism rotate at the same speed, thereby achieving an effect of bilateral synchronous movement in the electronic device, and preventing the flexible display panel main body from being pressed or pulled during folding or unfolding processes of the electronic device.

In summary, although the preferred embodiments of the present disclosure are disclosed as above, the above pre-

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ferred embodiments are not intended to limit the present disclosure. Those skilled in the art can make various changes and modifications without departing from the spirit and scope of the present disclosure. Therefore, the protection scope of the present disclosure is subject to the appended claims.

What is claimed is:

1. A hinge, comprising:
 - a fixed support;
 - a first rotation mechanism including a first rotation member and a first transmission member, wherein the first rotation member is rotatably connected to the fixed support, and the first transmission member is fixedly connected to the first rotation member;
 - a second rotation mechanism disposed opposite the first rotation mechanism and including a second rotation member and a second transmission member, wherein the second rotation member is rotatably connected to the fixed support, and the second transmission member is fixedly connected to the second rotation member; and
 - a moving block, wherein the first transmission member and the second transmission member are rotatably connected to the moving block, so as to drive the moving block away from or close to the fixed support; wherein the first transmission member includes a first link, the second transmission member includes a second link, and the moving block includes a base portion and a first extension portion and a second extension portion, which are connected to opposite sides of the base portion, wherein each of the first extension portion and the second extension portion includes two opposite extension plates extending from the base portion;
 - wherein one end of the first link is fixedly connected to the first rotation member, the other end of the first link is rotatably connected between the two extension plates of the first extension portion of the moving block, one end of the second link is fixedly connected to the second rotation member, and the other end of the second link is rotatably connected between the two extension plates of the second extension portion of the moving block.
2. The hinge of claim 1, wherein a first chute is defined by the fixed support, and the moving block is slidably disposed in the first chute.
3. The hinge of claim 2, wherein a wall surface of the first chute is provided with a protrusion, a groove is defined by the base portion of the moving block, and the protrusion is slidably received in the groove.
4. The hinge of claim 1, wherein the hinge further comprises a holding mechanism, and the holding mechanism includes a first holding arm, a second holding arm, and a first holding member, wherein the first holding member is provided with a first protrusion and a second protrusion, a first groove is defined by the first holding arm, and a second groove is defined by the second holding arm;
 - wherein the first holding arm is connected to the first rotation member, the second holding arm is connected to the second rotation member, and the first rotation member and the second rotation member are rotatably connected to the first holding member;
 - in response to the hinge being in one of a folded state or an unfolded state, the first protrusion is received in the first groove, and the second protrusion is received in the second groove; in response to the hinge being in the other state of the folded state or the unfolded state, the

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- first protrusion is separated from the first groove, and the second protrusion is separated from the second groove.
- 5. The hinge of claim 4, wherein the holding mechanism includes a first elastic member and a second elastic member, and the fixed support includes a main body portion and a first connection portion and a second connection portion, which extend from opposite sides of the main body portion,
 - wherein the first rotation member passes through the first connection portion, and the second rotation member passes through the second connection portion;
 - wherein the first elastic member surrounds the first rotation member and is clamped between the first holding member and the first connection portion, and the second elastic member surrounds the second rotation member and is clamped between the first holding member and the first connection portion.
- 6. The hinge of claim 4, wherein the hinge further comprises a first support plate and a second support plate; wherein the first support plate is connected to the first holding arm, so as to move under driving of the first holding arm, and the second support plate is connected to the second holding arm, so as to move under driving of the second holding arm.
- 7. The hinge of claim 6, wherein the hinge further comprises a first fixing member and a second fixing member, wherein a second chute is defined by the first support plate, a third chute is defined by the second support plate, the first fixing member passes through the second chute and is fixedly connected to the first holding arm, and the second fixing member passes through the third chute and is fixedly connected to the second holding arm;
 - wherein a position of the first fixing member in the second chute where the hinge is in the unfolded state is different from a position of the first fixing member in the second chute where the hinge being in the folded state; and a position of the second fixing member in the third chute where the hinge is in the unfolded state is different from a position of the second fixing member in the third chute where the hinge being in the folded state.
- 8. A flexible display panel, comprising:
 - a flexible display panel main body including a flexible portion and a first portion and a second portion, which are connected to opposite ends of the flexible portion; and
 - a hinge, including:
 - a fixed support;
 - a first rotation mechanism including a first rotation member and a first transmission member, wherein the first rotation member is rotatably connected to the fixed support, and the first transmission member is fixedly connected to the first rotation member;
 - a second rotation mechanism disposed opposite the first rotation mechanism and including a second rotation member and a second transmission member, wherein the second rotation member is rotatably connected to the fixed support, and the second transmission member is fixedly connected to the second rotation member; and
 - a moving block, wherein the first transmission member and the second transmission member are rotatably connected to the moving block, so as to drive the moving block away from or close to the fixed support; wherein the first portion is fixedly connected to the first rotation member, the second por-

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tion is fixedly connected to the second rotation member, and the moving block is located under the flexible portion;

wherein the first transmission member includes a first link, the second transmission member includes a second link, and the moving block includes a base portion and a first extension portion and a second extension portion, which are connected to opposite sides of the base portion, wherein each of the first extension portion and the second extension portion includes two opposite extension plates extending from the base portion;

wherein one end of the first link is fixedly connected to the first rotation member, the other end of the first link is rotatably connected between the two extension plates of the first extension portion of the moving block, one end of the second link is fixedly connected to the second rotation member, and the other end of the second link is rotatably connected between the two extension plates of the second extension portion of the moving block.

9. The flexible display panel of claim 8, wherein a first chute is defined by the fixed support, and the moving block is slidably disposed in the first chute.

10. The flexible display panel of claim 9, wherein a wall surface of the first chute is provided with a protrusion, a groove is defined by the base portion of the moving block, and the protrusion is slidably received in the groove.

11. The flexible display panel of claim 8, wherein the hinge further comprises a holding mechanism, and the holding mechanism includes a first holding arm, a second holding arm, and a first holding member, wherein the first holding member is provided with a first protrusion and a second protrusion, a first groove is defined by the first holding arm, and a second groove is defined by the second holding arm;

wherein the first holding arm is connected to the first rotation member, the second holding arm is connected to the second rotation member, and the first rotation member and the second rotation member are rotatably connected to the first holding member;

in response to the hinge being in one of a folded state or an unfolded state, the first protrusion is received in the first groove, and the second protrusion is received in the second groove; in response to the hinge being in the other state of the folded state or the unfolded state, the first protrusion is separated from the first groove, and the second protrusion is separated from the second groove.

12. The flexible display panel of claim 11, wherein the holding mechanism includes a first elastic member and a second elastic member, and the fixed support includes a main body portion and a first connection portion and a second connection portion, which extend from opposite sides of the main body portion, wherein the first rotation member passes through the first connection portion, and the second rotation member passes through the second connection portion;

wherein the first elastic member surrounds the first rotation member and is clamped between the first holding member and the first connection portion, and the second elastic member surrounds the second rotation member and is clamped between the first holding member and the first connection portion.

13. The flexible display panel of claim 11, wherein the hinge further comprises a first support plate and a second support plate;

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wherein the first support plate is connected to the first holding arm, so as to move under driving of the first holding arm, and the second support plate is connected to the second holding arm, so as to move under driving of the second holding arm.

14. The flexible display panel of claim 13, wherein the hinge further comprises a first fixing member and a second fixing member, wherein a second chute is defined by the first support plate, a third chute is defined by the second support plate, the first fixing member passes through the second chute and is fixedly connected to the first holding arm, and the second fixing member passes through the third chute and is fixedly connected to the second holding arm;

wherein a position of the first fixing member in the second chute where the hinge is in the unfolded state, is different from a position of the first fixing member in the second chute where the hinge is in the folded state; and a position of the second fixing member in the third chute where the hinge is in the unfolded state, is different from a position of the second fixing member in the third chute where the hinge is in the folded state.

15. An electronic device, comprising: a flexible display panel, and the flexible display panel includes:

a flexible display panel main body including a flexible portion and a first portion and

a second portion, which are connected to opposite ends of the flexible portion; and

a hinge, including:

a fixed support;

a first rotation mechanism including a first rotation member and a first transmission member, wherein the first rotation member is rotatably connected to the fixed support, and the first transmission member is fixedly connected to the first rotation member;

a second rotation mechanism disposed opposite the first rotation mechanism and including a second rotation member and a second transmission member, wherein the second rotation member is rotatably connected to the fixed support, and the second transmission member is fixedly connected to the second rotation member; and

a moving block, wherein the first transmission member and the second transmission member are rotatably connected to the moving block, so as to drive the moving block away from or close to the fixed support;

wherein the first portion is fixedly connected to the first rotation member, the second portion is fixedly connected to the second rotation member, and the moving block is located under the flexible portion;

wherein the first transmission member includes a first link, the second transmission member includes a second link, and the moving block includes a base portion and a first extension portion and a second extension portion, which are connected to opposite sides of the base portion, wherein each of the first extension portion and the second extension portion includes two opposite extension plates extending from the base portion;

wherein one end of the first link is fixedly connected to the first rotation member, the other end of the first link is rotatably connected between the two extension plates of the first extension portion of the moving block, one end of the second link is fixedly connected to the second rotation member, and the other end of the second link is rotatably connected

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between the two extension plates of the second extension portion of the moving block.

16. The electronic device of claim **15**, wherein a first chute is defined by the fixed support, and the moving block is slidably disposed in the first chute.

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17. The electronic device of claim **16**, wherein a wall surface of the first chute is provided with a protrusion, a groove is defined by the base portion of the moving block, and the protrusion is slidably received in the groove.

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